ASSIGNMENT 1

SOLUTIONS BY VIDHAN AGARWAL

BTECH

CSE

Ist YEAR

1.Write a C program to input two numbers and display their sum.

```
#include <stdio.h>
int main()
{
  int a,b,sum;
  printf("enter a number:");
  scanf("%d",&a);
  printf("enter a number:");
  scanf("%d",&b);
  printf("the sum of two numbers is %d",a+b);
  return 0;

}
```

2.Write a C program to perform addition, subtraction, multiplication, division, and modulus operations on two numbers.

```
#include <stdio.h>
int main()
{
   int a,b;
   printf("enter a number:");
   scanf("%d",&a);
   printf("enter a number:");
   scanf("%d",&b);
   printf("the sum of two numbers is %d \n",a+b);
   printf("the subtraction of two numbers is %d \n",a-b);
   printf("the product  of two numbers is %d \n",a*b);
   printf("the division of two numbers is %d \n",a/b);
   printf("the modulus of two numbers is %d \n",a%b);
   return 0;

}
```


3.Write a C program to calculate the area of a rectangle using length and breadth entered by the user.

```
#include <stdio.h>
int main()
{
   int a,b;
   printf("enter length of the rectangle:");
```

```
   scanf("%d",&a);

   printf("enter breadth of the rectangle:");

   scanf("%d",&b);

   printf("THE AREA OF THE RECTANGLE IS %d",a*b);


   return 0;


}
```

4.Write a C program to calculate the area of a circle using the radius entered by the user.

```
#include <stdio.h>

int main()

{

   int a;

   printf("enter radius of the circle:");

   scanf("%d",&a);


   printf("THE AREA OF THE circle IS %.2f",3.14*a*a);


   return 0;


}
```


5.Write a C program to swap two numbers using the third variable.

```
#include <stdio.h>

int main()
```

```
{
   int a ,b,c;
   printf("enter first number:");
   scanf("%d",&a);
    printf("enter second number:");
   scanf("%d",&b);
   printf(" THE TWO NUMBERS GIVEN BY USER ARE %d AND %d \n",a,b );
   c=a;
   a=b;
   b=c;
   printf("THE TWO NUMBERS AFTER SWAPPING ARE %d AND %d",a,b );
   return 0;
}
```


6.Write a C program to swap two numbers without using a third variable.

```
#include <stdio.h>
int main()
{
   int a ,b,c;
   printf("enter first number:");
   scanf("%d",&a);
    printf("enter second number:");
   scanf("%d",&b);
   printf(" THE TWO NUMBERS GIVEN BY USER ARE %d AND %d \n",a,b );
   a=a+b;
   b=a-b;
   a=a-b;
```

```
    printf("THE TWO NUMBERS AFTER SWAPPING ARE %d AND %d",a,b );
    return 0;
}
```

7.Write a C program to check whether a number is even or odd.

```
#include <stdio.h>
int main()
{
    int a ;
    printf("enter  number:");
    scanf("%d",&a);
    if(a%2==0)
    {
      printf("%d is even",a);

    }
    else
    printf("%d is odd",a);


return 0;
}
```


8.Write a C program to check whether a number is positive, negative, or zero.

```
#include <stdio.h>
int main()
{
```

```
   int a ;
   printf("enter  number:");
   scanf("%d",&a);
   if(a<0)
   {
      printf("%d is negative",a);

   }
   else if(a>0)
   printf("%d is positive",a);
   else
   printf("%d is zero",a);
return 0;
}
```


9.Write a C program to find the largest among two numbers.

```
int main()
{
   int a,b ;
   printf("enter first  number:");
   scanf("%d",&a);
   printf("enter second  number:");
   scanf("%d",&b);
   if(a<b)
   printf("%d is largest",b);
   else
   printf("%d is greatest",a);
```

```
return 0;

}
```

10. Write a C program to find the largest among three numbers.

```
#include <stdio.h>

int main()

{

  int a,b,c ;

  printf("enter first  number:");

  scanf("%d",&a);

  printf("enter second  number:");

  scanf("%d",&b);

  printf("enter third  number:");

  scanf("%d",&c);

  if(a<b && c<b)

  printf("%d is largest",b);

  else if(b<a && c<a)

  printf("%d is greatest",a);

  else

  printf("%d is greatest",c);

return 0;

}
```

11.Write a C program to check whether a person is eligible for voting.

```
#include <stdio.h>
int main()
{
   int a ;
   printf("enter your age:");
   scanf("%d",&a);

   if(a<18)
   printf("you are not eligible for voting");
   else
   printf("you are  eligible for voting");
return 0;
}
```


12.Write a C program to check whether a given year is a leap year.

```
#include <stdio.h>
int main()
{
   int a ;
   printf("enter the year:");
   scanf("%d",&a);
   if(a%4==0)
   printf(" %d is leap year",a);
   else
   printf(" %d is not leap year",a);
```

```
return 0;
}
```

13.Write a C program to check whether an alphabet is a vowel or consonant.

```
#include <stdio.h>
int main()
{
  char a ;
  printf("enter the alphabet:");
  scanf("%C",&a);
  if(a == 'a' || a == 'e' || a == 'i' || a == 'o' || a == 'u' ||
    a == 'A' || a == 'E' || a == 'I' || a == 'O' || a == 'U')
  printf(" %C is vowel",a);
  else
  printf(" %C is consonant",a);
return 0;
}
```


14.Write a C program to check whether a number is divisible by both 5 and 11.

```
#include <stdio.h>
int main()
{
  int a ;
  printf("enter the num:");
  scanf("%d",&a);
  if(a%55==0)
```

```
  printf(" %d is divisible by both 5 and 11",a);
  else
  printf(" %d is not divisible by both 5 and 11",a);
return 0;
}
```

15. Write a C program to calculate the grade of a student according to marks obtained.

```
#include <stdio.h>
int main()
{
  int a ;
  printf("enter the perrcentage of marks:");
  scanf("%d",&a);
  if(a>91)
  printf("A GRADE");
  else if(a<=91 && a>81)
  printf("B GRADE");
  else if(a<=81 && a>71)
  printf("C GRADE");
  else if(a<=71 && a>61)
  printf("D GRADE");
  else
  printf("E GRADE");
return 0;
}
```


16. Write a C program to display the day name according to the day number using switch case.

```
#include <stdio.h>
```

```
int main()

{
  int a ;
  printf("enter the day number(1-7):");
  scanf("%d",&a);
  switch(a){
    case 1:
    printf("MONDAY");
    break;
     case 2:
    printf("TUESDAY");
    break;
     case 3:
    printf("WEDNESDAY");
    break;
     case 4:
    printf("THURSDAY");
    break;
     case 5:
    printf("FRIDAY");
    break;
     case 6:
    printf("SATURDAY");
    break;
    default:
    printf("SUNDAY");
    }
```

```
return 0;

}
```

17.Write a C program to display the month name according to the month number using switch case.

```
int main()


{

  int a ;

  printf("enter the month number(1-12):");

  scanf("%d",&a);

  switch(a){

    case 1:

    printf("JAN");

    break;

     case 2:

    printf("FEB");

    break;

     case 3:

    printf("MAR");

    break;

     case 4:

    printf("APR");

    break;

     case 5:

    printf("MAY");

    break;

     case 6:
```

```
printf("JUNE");

break;

case 7:

printf("JULY");

break;

case 8:

printf("AUG");

break;

case 9:

printf("SEP");

break;

case 10 :

printf("OCT");

break;

case 11:

printf("NOV");

break;

default:

printf("DEC");

}

return 0;

}
```


18.Write a C program to calculate electricity bill according to units consumed.

```
#include <stdio.h>


int main() {
```

```
    int units;
    float bill = 0;

    printf("Enter units consumed: ");
    scanf("%d", &units);

    if (units <= 200) {
        bill = units * 3.0;
    } else if (units <= 400) {
        bill = (200 * 3.0) + (units - 200) * 5.0;
    } else if (units <= 800) {
        bill = (200 * 3.0) + (200 * 5.0) + (units - 400) * 6.5;
    } else if (units <= 1200) {
        bill = (200 * 3.0) + (200 * 5.0) + (400 * 6.5) + (units - 800) * 7.0;
    } else {
        bill = (200 * 3.0) + (200 * 5.0) + (400 * 6.5) + (400 * 7.0) + (units - 1200) * 8.0;
    }

    printf("Electricity Bill = Rs. %.2f\n", bill);
    return 0;
}
```


19. Write a C program to calculate income tax according to income slabs.

```
#include <stdio.h>

int main() {
```

```
    float income, tax = 0;

    printf("Enter your annual income: ");
    scanf("%f", &income);

    if (income <= 250000) {
        tax = 0;   // No tax up to ₹2.5 lakh
    } else if (income <= 500000) {
        tax = (income - 250000) * 0.05;
    } else if (income <= 1000000) {
        tax = (250000 * 0.05) + (income - 500000) * 0.20;
    } else {
        tax = (250000 * 0.05) + (500000 * 0.20) + (income - 1000000) * 0.30;
    }
```



```
    // Add 4% health & education cess
    tax = tax + (tax * 0.04);

    printf("Income Tax Payable = Rs. %.2f\n", tax);

    return 0;
}
```

20.Write a C program to calculate profit or loss based on cost price and selling price.

```
#include <stdio.h>

int main() {
```

```
    float cp, sp;

    printf("Enter Cost Price: ");
    scanf("%f", &cp);

    printf("Enter Selling Price: ");
    scanf("%f", &sp);

    if (sp > cp) {
        printf("Profit = %.2f\n", sp - cp);
    } else if (cp > sp) {
        printf("Loss = %.2f\n", cp - sp);
    } else {
        printf("No Profit, No Loss\n");
    }

    return 0;
}
```


21.Write a C program to check whether a character is an alphabet, digit, or special character.

```
#include <stdio.h>

int main() {
    char ch;

    printf("Enter a character: ");
    scanf("%c", &ch);
```

```
    if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
        printf("It is an Alphabet\n");
    } else if (ch >= '0' && ch <= '9') {
        printf("It is a Digit\n");
    } else {
        printf("It is a Special Character\n");
    }

    return 0;
}
```

22. Write a C program to find the greatest among four numbers.


```
#include <stdio.h>

int main() {
    int a, b, c, d;

    printf("Enter four numbers: ");
    scanf("%d %d %d %d", &a, &b, &c, &d);

    if (a >= b && a >= c && a >= d) {
        printf("Greatest = %d\n", a);
    } else if (b >= a && b >= c && b >= d) {
        printf("Greatest = %d\n", b);
    } else if (c >= a && c >= b && c >= d) {
        printf("Greatest = %d\n", c);
```

```
    } else {
        printf("Greatest = %d\n", d);
    }

    return 0;
}
```

23. Write a C program to calculate total marks of five subject and display all subject marks also, percentage, and division of a student.


```
#include <stdio.h>

int main() {
    int s1, s2, s3, s4, s5, total;
    float percentage;

    printf("Enter marks of 5 subjects: ");
    scanf("%d %d %d %d %d", &s1, &s2, &s3, &s4, &s5);

    total = s1 + s2 + s3 + s4 + s5;
    percentage = total / 5.0;

    printf("\nMarks in each subject:\n");
    printf("Subject 1 = %d\n", s1);
    printf("Subject 2 = %d\n", s2);
    printf("Subject 3 = %d\n", s3);
    printf("Subject 4 = %d\n", s4);
    printf("Subject 5 = %d\n", s5);
```

```c
    printf("\nTotal Marks = %d\n", total);
    printf("Percentage = %.2f\n", percentage);

    if (percentage >= 60) {
        printf("Division = First\n");
    } else if (percentage >= 45) {
        printf("Division = Second\n");
    } else if (percentage >= 33) {
        printf("Division = Third\n");
    } else {
        printf("Division = Fail\n");
    }
```



```c
    return 0;
}
```

24. Write a C program to check whether three sides can form a valid triangle.

```c
#include <stdio.h>

int main() {
    int a, b, c;

    printf("Enter three sides of a triangle: ");
    scanf("%d %d %d", &a, &b, &c);

    if (a + b > c && a + c > b && b + c > a) {
```

```
        printf("Valid Triangle\n");
    } else {
        printf("Not a Valid Triangle\n");
    }

    return 0;
}
```

25. Write a C program to check whether a character is uppercase or lowercase.

```
#include <stdio.h>

int main() {
    char ch;
```



```
    printf("Enter a character: ");
    scanf("%c", &ch);

    if (ch >= 'A' && ch <= 'Z') {
        printf("Uppercase Letter\n");
    } else if (ch >= 'a' && ch <= 'z') {
        printf("Lowercase Letter\n");
    } else {
        printf("Not an Alphabet Letter\n");
    }

    return 0;
}
```

26. Write a C program to calculate the square and cube of a number.

```
#include <stdio.h>

int main() {
  int num, square, cube;

  printf("Enter a number: ");
  scanf("%d", &num);

  square = num * num;
  cube = num * num * num;

  printf("Square = %d\n", square);
  printf("Cube = %d\n", cube);

  return 0;
}
```


27. Write a C program to check whether a given number is an Armstrong number.

```
#include <stdio.h>

int main() {
  int num, sum = 0, temp, r;

  printf("Enter a number: ");
  scanf("%d", &num);
```

```
    temp = num;

    while (temp > 0) {
        r = temp % 10;          // get last digit
        sum = sum + (r * r * r);   // cube and add
        temp = temp / 10;        // remove last digit
    }

    if (sum == num)
        printf("Armstrong Number\n");
    else
        printf("Not an Armstrong Number\n");

    return 0;
}
```


28. Write a C program to check whether a given number is a Palindrome number.

```
#include <stdio.h>

int main() {
    int num, reversed = 0, remainder, original;

    printf("Enter a number: ");
    scanf("%d", &num);

    original = num;
```

```
  while (num > 0) {
    remainder = num % 10;           // get last digit
    reversed = reversed * 10 + remainder; // build reversed number
    num = num / 10;               // remove last digit
  }

  if (original == reversed)
    printf("Palindrome Number\n");
  else
    printf("Not a Palindrome Number\n");

  return 0;
}
```


29.Write a C program to calculate Simple Interest.

```
#include <stdio.h>

int main() {
  float principal, rate, time, si;

  printf("Enter Principal amount: ");
  scanf("%f", &principal);

  printf("Enter Rate of interest: ");
  scanf("%f", &rate);

  printf("Enter Time (in years): ");
```

```
    scanf("%f", &time);

    si = (principal * rate * time) / 100;

    printf("Simple Interest = %.2f\n", si);

    return 0;
}
```

30.Write a C program to calculate Compound Interest.

```
#include <stdio.h>
#include <math.h>

int main() {
    float principal, rate, time, ci;

    printf("Enter Principal amount: ");
    scanf("%f", &principal);

    printf("Enter Rate of interest: ");
    scanf("%f", &rate);

    printf("Enter Time (in years): ");
    scanf("%f", &time);

    ci = principal * (pow((1 + rate / 100), time)) - principal;

    printf("Compound Interest = %.2f\n", ci);
```

```
    return 0;
}
```

31. Write a C program to calculate the Gross Salary of an employee.

```
#include <stdio.h>

int main() {
    float basic, hra, da, gross;

    printf("Enter Basic Salary: ");
    scanf("%f", &basic);

    // Assume HRA)(HOUSE RENT ALLOWANCE) = 20% of basic, DA(DEARNESS
ALLOWANCE = 40% of basic
    hra = 0.20 * basic;
    da  = 0.40 * basic;

    gross = basic + hra + da;

    printf("Basic Salary = %.2f\n", basic);
    printf("HRA = %.2f\n", hra);
    printf("DA = %.2f\n", da);
    printf("Gross Salary = %.2f\n", gross);

    return 0;
}
```

32. Write a C program to convert temperature from Celsius to Fahrenheit.

```
#include <stdio.h>

int main() {
   float celsius, fahrenheit;

   printf("Enter temperature in Celsius: ");
   scanf("%f", &celsius);

   fahrenheit = (celsius * 9 / 5) + 32;

   printf("Temperature in Fahrenheit = %.2f\n", fahrenheit);

   return 0;
}
```


33. Write a C program to calculate sales commission according to sales amount.

```
#include <stdio.h>
int main() {
   float sales, commission;
   printf("Enter sales amount: ");
   scanf("%f", &sales);

   if (sales <= 5000)
      commission = sales * 0.02;
   else if (sales <= 10000)
```

```
    commission = sales * 0.05;
  else
    commission = sales * 0.10;

  printf("Commission = %.2f\n", commission);
  return 0;
}
```

34. Write a C program to find the larger number between two numbers using the ternary operator.

```
#include <stdio.h>
int main() {
  int a, b, larger;
  printf("Enter two numbers: ");
  scanf("%d %d", &a, &b);

  larger = (a > b) ? a : b;
  printf("Larger number is %d\n", larger);
  return 0;
}
```


35. Write a C program to check whether a number is even or odd using the ternary operator.

```
#include <stdio.h>
int main() {
  int num;
  printf("Enter a number: ");
  scanf("%d", &num);
```

```
    (num % 2 == 0) ? printf("Even\n") : printf("Odd\n");
    return 0;
}
```

36.Write a C program to find the absolute value of a number.

```
#include <stdio.h>
int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

    if (num < 0)
        num = -num;

    printf("Absolute value = %d\n", num);
    return 0;
}
```


37.Write a C program to create a menu-driven banking system using switch case.

```
#include <stdio.h>
int main() {
    int choice;
    float balance = 1000, amount;

    while (1) {
```

```
printf("\n--- Banking Menu ---\n");

printf("1. Deposit\n2. Withdraw\n3. Check Balance\n4. Exit\n");

printf("Enter your choice: ");

scanf("%d", &choice);


switch (choice) {

   case 1:

      printf("Enter amount to deposit: ");

      scanf("%f", &amount);

      balance += amount;

      printf("Deposited successfully!\n");

      break;

   case 2:

      printf("Enter amount to withdraw: ");

      scanf("%f", &amount);

      if (amount <= balance) {

         balance -= amount;

         printf("Withdrawn successfully!\n");

      } else {

         printf("Insufficient balance!\n");

      }

      break;

   case 3:

      printf("Current Balance = %.2f\n", balance);

      break;

   case 4:

      return 0;

   default:
```

```
            printf("Invalid choice!\n");
        }
    }
}
```

38. Write a C program to create a student result system that displays all subject marks, total marks, percentage, and result.


```
#include <stdio.h>
int main() {
    int marks[5], total = 0;
    float percentage;
    char *result;

    printf("Enter marks of 5 subjects: ");
    for (int i = 0; i < 5; i++) {
        scanf("%d", &marks[i]);
        total += marks[i];
    }

    percentage = total / 5.0;
    result = (percentage >= 40) ? "Pass" : "Fail";

    printf("\n--- Result ---\n");
    for (int i = 0; i < 5; i++)
        printf("Subject %d: %d\n", i + 1, marks[i]);

    printf("Total Marks = %d\n", total);
```

```
   printf("Percentage = %.2f\n", percentage);
   printf("Result = %s\n", result);

   return 0;
}
```

39. Write a C program to display the season according to the month number using switch case.

```
#include <stdio.h>
int main() {
   int month;
   printf("Enter month number (1-12): ");
   scanf("%d", &month);

   switch (month) {
      case 12: case 1: case 2:
         printf("Season: Winter\n");
         break;
      case 3: case 4: case 5:
         printf("Season: Spring\n");
         break;
      case 6: case 7: case 8:
         printf("Season: Summer\n");
         break;
      case 9: case 10: case 11:
         printf("Season: Autumn\n");
         break;
      default:
```

```
        printf("Invalid month!\n");
    }
    return 0;
}
```

40. Write a C program to verify username and password entered by the user.

```
#include <stdio.h>
#include <string.h>

int main() {
    char username[20], password[20];

    printf("Enter username: ");
    scanf("%s", username);

    printf("Enter password: ");
    scanf("%s", password);

    if (strcmp(username, "admin") == 0 && strcmp(password, "1234") == 0) {
        printf("Login successful!\n");
    } else {
        printf("Login failed!\n");
    }

    return 0;
}
```

41.Write a C program to check whether an alphabet is a vowel or consonant using switch case.

```
#include <stdio.h>
int main() {
  char ch;
  printf("Enter an alphabet: ");
  scanf(" %c", &ch);

  switch (ch) {
    case 'a': case 'e': case 'i': case 'o': case 'u':
    case 'A': case 'E': case 'I': case 'O': case 'U':
      printf("Vowel\n");
      break;
    default:
      printf("Consonant\n");
  }
  return 0;
}
```


42.Write a C program to create an advanced calculator using switch case.

```
#include <stdio.h>
int main() {
  int choice;
  float a, b;
  printf("Enter two numbers: ");
  scanf("%f %f", &a, &b);
```

```
  printf("\n--- Calculator Menu ---\n");
  printf("1. Addition\n2. Subtraction\n3. Multiplication\n4. Division\n");
  printf("Enter your choice: ");
  scanf("%d", &choice);

  switch (choice) {
    case 1: printf("Result = %.2f\n", a + b); break;
    case 2: printf("Result = %.2f\n", a - b); break;
    case 3: printf("Result = %.2f\n", a * b); break;
    case 4:
      if (b != 0) printf("Result = %.2f\n", a / b);
      else printf("Division by zero not allowed!\n");
      break;
    default: printf("Invalid choice!\n");
  }
  return 0;
}
```


43. Write a C program to check whether a number is divisible by 2, 3, and 5.

```
#include <stdio.h>
int main() {
  int num;
  printf("Enter a number: ");
  scanf("%d", &num);

  if (num % 2 == 0 && num % 3 == 0 && num % 5 == 0)
    printf("%d is divisible by 2, 3, and 5\n", num);
```

```
  else
    printf("%d is not divisible by 2, 3, and 5\n", num);

  return 0;
}
```

44. Write a C program to create a simple ATM system using switch case.

```
#include <stdio.h>
int main() {
  int choice;
  float balance = 5000, amount;

  while (1) {
    printf("\n--- ATM Menu ---\n");
    printf("1. Deposit\n2. Withdraw\n3. Check Balance\n4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
      case 1:
        printf("Enter amount to deposit: ");
        scanf("%f", &amount);
        balance += amount;
        printf("Deposited successfully!\n");
        break;
      case 2:
        printf("Enter amount to withdraw: ");
```

```
            scanf("%f", &amount);
            if (amount <= balance) {
              balance -= amount;
              printf("Withdrawn successfully!\n");
            } else {
              printf("Insufficient balance!\n");
            }
            break;
          case 3:
            printf("Current Balance = %.2f\n", balance);
            break;
          case 4:
            return 0;
          default:
            printf("Invalid choice!\n");
        }
    }
}
```


45. Write a C program to calculate employee bonus according to salary.

```
#include <stdio.h>
int main() {
  float salary, bonus;
  printf("Enter employee salary: ");
  scanf("%f", &salary);

  if (salary <= 5000)
```

```
    bonus = salary * 0.10;
  else if (salary <= 10000)
    bonus = salary * 0.08;
  else
    bonus = salary * 0.05;

  printf("Bonus = %.2f\n", bonus);
  return 0;
}
```

46. Write a C program to calculate mobile bill according to call minutes.


```
#include <stdio.h>
int main() {
  int minutes;
  float bill;
  printf("Enter call minutes: ");
  scanf("%d", &minutes);

  if (minutes <= 100)
    bill = minutes * 1.0;   // Rs.1 per minute
  else if (minutes <= 200)
    bill = minutes * 0.80;  // Rs.0.80 per minute
  else
    bill = minutes * 0.50;  // Rs.0.50 per minute

  printf("Mobile Bill = %.2f\n", bill);
  return 0;
```

```
}
```

47. Write a C program to calculate railway fare according to passenger age.

```
#include <stdio.h>
int main() {
   int age;
   float fare = 100; // base fare
   printf("Enter passenger age: ");
   scanf("%d", &age);

   if (age < 5)
      printf("No fare for children under 5\n");
   else if (age <= 12)
      printf("Fare = %.2f\n", fare * 0.5); // half fare
   else if (age >= 60)
      printf("Fare = %.2f\n", fare * 0.7); // 30% discount
   else
      printf("Fare = %.2f\n", fare);

   return 0;
}
```


48. Write a C program to calculate Body Mass Index (BMI).

```
#include <stdio.h>
int main() {
   float weight, height, bmi;
```

```
    printf("Enter weight (kg): ");
    scanf("%f", &weight);
    printf("Enter height (m): ");
    scanf("%f", &height);

    bmi = weight / (height * height);
    printf("BMI = %.2f\n", bmi);

    if (bmi < 18.5)
        printf("Underweight\n");
    else if (bmi < 25)
        printf("Normal\n");
    else if (bmi < 30)
        printf("Overweight\n");
    else
        printf("Obese\n");

    return 0;
}
```


49. Write a C program to determine scholarship eligibility based on percentage obtained.

```
#include <stdio.h>
int main() {
    float percentage;
    printf("Enter percentage: ");
    scanf("%f", &percentage);
```

```
    if (percentage >= 75)
        printf("Eligible for scholarship\n");
    else
        printf("Not eligible for scholarship\n");

    return 0;
}
```

50. Write a C program to find the smallest among three numbers.

```
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    if (a <= b && a <= c)
        printf("Smallest = %d\n", a);
    else if (b <= a && b <= c)
        printf("Smallest = %d\n", b);
    else
        printf("Smallest = %d\n", c);

    return 0;
}
```

51. Write a C program to find the average of five numbers.

```
#include <stdio.h>
int main() {
   int n1, n2, n3, n4, n5;
   float avg;
   printf("Enter five numbers: ");
   scanf("%d %d %d %d %d", &n1, &n2, &n3, &n4, &n5);

   avg = (n1 + n2 + n3 + n4 + n5) / 5.0;
   printf("Average = %.2f\n", avg);

   return 0;
}
```


52. Write a C program to calculate the perimeter of a rectangle. #include <stdio.h>

```
int main() {
   float length, width, perimeter;
   printf("Enter length and width: ");
   scanf("%f %f", &length, &width);

   perimeter = 2 * (length + width);
   printf("Perimeter = %.2f\n", perimeter);

   return 0;
}
```

53. Write a C program to calculate the circumference of a circle.

```
#include <stdio.h>

int main() {
   float radius, circumference;
   printf("Enter radius: ");
   scanf("%f", &radius);

   circumference = 2 * 3.14* radius;
   printf("Circumference = %.2f\n", circumference);

   return 0;
}
```


54. Write a C program to check whether a number is divisible by 7.

```
#include <stdio.h>
int main() {
   int num;
   printf("Enter a number: ");
   scanf("%d", &num);

   if (num % 7 == 0)
      printf("%d is divisible by 7\n", num);
   else
      printf("%d is not divisible by 7\n", num);
```

```
    return 0;
}
```

55. Write a C program to calculate the area of a triangle.

```
#include <stdio.h>
int main() {
    float base, height, area;
    printf("Enter base and height: ");
    scanf("%f %f", &base, &height);

    area = 0.5 * base * height;
    printf("Area of triangle = %.2f\n", area);

    return 0;
}
```